

ULTRASOUND DATA PROCESSING

Ultrasound video conversion (.tvd to .mat)

1. You will need to install Echo Wave II on your computer. This can be downloaded from the Telemed website <https://www.pcultrasound.com/support/software/>
2. After installing Echo Wave II, you will need to go to the installation folder "C:\Program Files (x86)\Telemed\Echo Wave II\Config\Options" → open "options2.txt" → change the "0" in line 10: "disable_hardware_configuration_checking; 0" to a 1 → save and close the file. This allows Echo Wave to open without the computer being connected to the ultrasound
3. Have downloaded the ultrasound conversion code from https://drive.google.com/drive/folders/1cuZf_-xLXYv_G0mRo2_uaEqBw49Vn5Ye?usp=drive_link
4. Run Echo Wave II as an administrator
5. Run Matlab as an administrator
6. With the ultrasound conversion files added to your Matlab path, open and run the "TVD_2_MAT.m" script
7. Select the .tvd ultrasound files that you want to convert and let the code run

For tracking relatively long or relatively dynamic tasks

Instructions for downloading and using the necessary software can be found at <https://github.com/timvanderzee/UltraTimTrack>. I highly recommend cloning the Git repository to your computer to avoid unexpected errors and to always have the most up to date version of the code

For tracking relatively short or relatively slow / controlled tasks

1. Clone https://github.com/brentrat/UltraTrack_v5_3 to your computer.
2. With the code added to your Matlab path, open and run the "UltraTrack_v5_3.m" script
3. Click "File" → "Open Video File" → select a converted ultrasound .mat file
4. Set the "Res. (mmPerPix)" to the correct value. For me, this value is 0.093 mm/Pix. To calculate this value, open Echo Wave II → load in your ultrasound .tvd file → use the measurement tool to record the width of the image (e.g. 52 mm) → load your converted ultrasound .mat file to your Matlab workspace → within the loaded structure, record the width of the image variable (e.g. 555 Pix) → divide mm by Pix (e.g. 52 / 555).
5. I recommend you now find a frame where the fascicles of the muscle of interest look to be at their shortest length. It can help to load in corresponding force or marker data from the same trial and limb from a .c3d file to help determine where you are in the movement cycle. This will help with selecting frames for "Keyframe Correction" in a later step as well: Click "File" → "Load Data File" → select the .c3d file of interest
6. Click "Define ROI" and select the corners of your region (muscle) of interest. Double-click to set the ROI.
7. Check the "Fixed ROI" box
8. Click "Define Fascicle" and click on the approximate origin and insertion of a fascicle
9. Click "Process All" and let the code run
10. Observe the tracking results and perform a keyframe correction to remove drift. To do this: choose a particular point in the movement cycle → at each frame in the video that corresponds to that exact point you will click "Add" under "Keyframe Correction" → once all points are added, click "Correct"

Keyframes". Again, to help with determining where in the movement cycle you are, it helps to have loaded in force or marker data

11. Save the results by Clicking "File" → "Save tracking data as" → "MAT"